THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · MEDIUM LEVEL MCQ TEST SET 2

Full Syllabus (Chapters 1–14) | RBSE Pattern | Session 2026–27

Time: 1 Hour Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–50 are Multiple Choice Questions (1 mark each); there is no negative marking.
3. Choose the correct option and write its letter (a, b, c or d) against the question number.
4. Questions cover the whole syllabus of the class.
5. Medium level: multi-step questions.

MULTIPLE CHOICE QUESTIONS — 50 × 1 = 50

Choose the correct option and write it in your answer book.

1. A storm breaks a tree at a height of 3 m above the ground. The broken part bends over, its top touches the ground and the broken part makes an angle of 30° with the ground. The distance from the foot of the tree to the point where its top touches the ground is: 1

(a) 6 m **(b)** 3 m **(c)** $3\sqrt{3}$ m **(d)** $\sqrt{3}$ m

2. Two concentric circles have radii 52 cm and 20 cm. A chord of the larger circle touches the smaller circle. The length of the chord is: 1

(a) 96 cm **(b)** 48 cm **(c)** 64 cm **(d)** 72 cm

3. Two concentric circles have radii 14 cm and 7 cm. A sector of angle 60° is marked in both. The area of the region between the two arcs within the sector is: 1

(a) 77 cm² **(b)** 462 cm² **(c)** 154 cm² **(d)** 38.5 cm²

4. A solid metallic sphere of radius 12 cm is melted and drawn into a wire of circular cross-section of radius 3 cm. The length of the wire is: 1

(a) 768 cm **(b)** 512 cm **(c)** 128 cm **(d)** 256 cm

5. The mean of a distribution with classes 10–20, 20–30, 30–40, 40–50, 50–60, 60–70 and frequencies 13, 4, p, 8, 14, 5 respectively is 39.2. The value of p is: 1

(a) 11 **(b)** 5 **(c)** 6 **(d)** 8

6. A bag contains 3 red balls and x blue balls. The probability of drawing a blue ball is 2 times the probability of drawing a red ball. The value of x is: 1

(a) 2 **(b)** 9 **(c)** 5 **(d)** 6

7. The decimal expansion of $\frac{77}{210}$ is: 1

(a) an integer **(b)** non-terminating non-repeating
(c) non-terminating repeating **(d)** terminating

8. If the zeroes of $x^2 + px + q$ are 1 and 6, then p equals: 1

(a) 6 **(b)** -7 **(c)** -6 **(d)** 7

9. If $3x + 2y = 25$ and $2x + 3y = 20$, then $x + y =$ 1

(a) 45 **(b)** 5 **(c)** $\frac{45}{2}$ **(d)** 9

10. The equation $x^2 + 6x + 8 = 0$ has: 1

(a) two equal real roots **(b)** three real roots
(c) two distinct real roots **(d)** no real roots

11. The middle term of an AP with 13 terms, first term 5 and last term 41, is: 1

(a) 20 **(b)** 41 **(c)** 26 **(d)** 23

12. The diagonals of a rhombus are 12 and 16. Its side squared is: 1

(a) 200 **(b)** 196 **(c)** 100 **(d)** 400

13. The ratio $k:1$ in which the line $x=5$ divides the join of $(2,0)$ and $(11,0)$ has $k=$ 1

(a) $\frac{1}{2}$ **(b)** $\frac{5}{6}$ **(c)** $\frac{1}{3}$ **(d)** 2

14. If $\tan\theta=\frac{4}{3}$, then $\dfrac{\sin\theta+\cos\theta}{\cos\theta}=$ 1

(a) $\frac{1}{3}$ **(b)** $\frac{3}{7}$ **(c)** $\frac{4}{3}$ **(d)** $\frac{7}{3}$

15. A storm breaks a tree at a height of 3 m above the ground. The broken part bends over, its top touches the ground and the broken part makes an angle of 60° with the ground. The original height of the tree is: 1

(a) $2\sqrt{3}$ m **(b)** 6 m **(c)** 5 m **(d)** $3+2\sqrt{3}$ m

16. In $\triangle ABC$, right-angled at B, $AB=11$ cm and $BC=60$ cm. A circle is inscribed in the triangle. The radius of the inscribed circle is: 1

(a) 6 cm **(b)** 30 cm **(c)** 5 cm **(d)** 10 cm

17. An arc of length 22 cm subtends an angle at the centre of a circle of radius 14 cm. The angle subtended by the arc at the centre is: 1

(a) 180° **(b)** 45° **(c)** 120° **(d)** 90°

18. A solid sphere of radius 3 cm is completely dropped into water in a cylindrical vessel of base radius 6 cm, and no water spills. The rise in the water level is: 1

(a) 1 cm **(b)** 0.5 cm **(c)** 2 cm **(d)** 3 cm

19. The median of a distribution with classes 60–70, 70–80, 80–90, 90–100, 100–110, 110–120 and frequencies 18, 18, x, 20, y, 12 respectively, and total frequency 100, is 91.5. The value of x is: 1

(a) 12 **(b)** 23 **(c)** 10 **(d)** 11

20. A bag contains 3 red and 5 blue balls. When x more red balls are added, the probability of drawing a red ball becomes $\frac{1}{2}$. The value of x is: 1

(a) 4 **(b)** 1 **(c)** 2 **(d)** 3

21. If $\sqrt{3}$ is irrational, then $\sqrt{3}\times\sqrt{3}$ is: 1

(a) not defined **(b)** irrational

(c) rational **(d)** non-terminating non-repeating

22. One zero of $x^2-7x+12$ is 3. The other zero is: 1

(a) 4 **(b)** 9 **(c)** -4 **(d)** 10

23. If $x+y=9$ and $x-y=3$, then $x^2-y^2=$ 1

(a) 72 **(b)** 90 **(c)** 27 **(d)** 12

24. The larger root of $x^2-11x+24=0$ is: 1

(a) 11 **(b)** 3 **(c)** 24 **(d)** 8

25. If $a_8=25$ and $d=3$, then $a=$ 1

(a) 4 **(b)** 1 **(c)** 7 **(d)** 46

26. Two poles of heights 13 m and 8 m are 12 m apart. The distance between their tops (m) is: 1

(a) 5 **(b)** 17 **(c)** 13 **(d)** 12

27. If $(1, 2)$, $(3, 6)$ and $(5, k)$ are collinear, then $k =$ 1

(a) 12 **(b)** 6 **(c)** 8 **(d)** 10

28. If $\sin 20^\circ = \cos B$ for an acute angle B, then $B =$ 1

(a) 20° **(b)** 45° **(c)** 160° **(d)** 70°

29. The string of a kite is 10 m long. It is taut and makes an angle of 60° with the level ground. The height of the kite above the ground is: 1

(a) $10\sqrt{3}$ m **(b)** $5\sqrt{3}$ m **(c)** $\frac{5\sqrt{3}}{2}$ m **(d)** 5 m

30. Tangents QA and QB are drawn from an external point Q to a circle, and $QA = (8x + 3)$ cm and $QB = (5x + 18)$ cm. The length of QA is: 1

(a) 5 cm **(b)** 43 cm **(c)** 58 cm **(d)** 45 cm

31. A chord of a circle of radius 49 cm subtends a right angle at the centre. The area of the minor segment is: 1

(a) 1286.25 cm² **(b)** 1886.5 cm² **(c)** 1200.5 cm² **(d)** 686 cm²

32. A capsule is a cylinder of radius 7 cm and length 10 cm (straight part) with a hemisphere of the same radius fixed at each end. The total surface area of the capsule is: 1

(a) 2112 cm² **(b)** 1364 cm² **(c)** 748 cm² **(d)** 1056 cm²

33. The median of a distribution with classes 0–10, 10–20, 20–30, 30–40, 40–50, 50–60, 60–70 and frequencies 21, 3, x, 10, y, 14, 9 respectively, and total frequency 100, is 35. The value of y is: 1

(a) 23 **(b)** 20 **(c)** 24 **(d)** 22

34. A red die and a blue die are rolled together. The probability that both numbers are even is: 1

(a) $\frac{3}{4}$ **(b)** $\frac{1}{4}$ **(c)** $\frac{2}{9}$ **(d)** $\frac{5}{18}$

35. The HCF of 45, 75 and 90 is: 1

(a) 15 **(b)** 450 **(c)** 5 **(d)** 30

36. If α, β are zeroes of $x^2 - 7x + 12$, then $\alpha^2 + \beta^2$ is: 1

(a) 25 **(b)** 73 **(c)** 37 **(d)** 49

37. The value of x in $5x + 3y = 29$, $4x + 7y = 37$ is: 1

(a) -4 **(b)** 3 **(c)** 4 **(d)** 7

38. The sum of the roots of $3x^2 + 7x - 6 = 0$ is: 1

(a) -2 **(b)** $-\frac{7}{3}$ **(c)** 2 **(d)** $\frac{7}{3}$

39. The 9th term of the AP whose 1st term is 4 and 4th term is 22 is: 1

(a) 46 **(b)** 54 **(c)** 58 **(d)** 52

40. If $DE \parallel BC$ in $\triangle ABC$, $AB = 7$ cm, $AD = 4$ cm, $AC = 21$ cm, then EC (cm) is: 1

(a) 12 **(b)** 3 **(c)** 9 **(d)** 21

41. The perimeter of the triangle with vertices $(0, 0)$, $(5, 0)$, $(0, 12)$ is: 1

(a) 43 **(b)** 30 **(c)** 34 **(d)** 17

42. The value of $2\sin^2 60^\circ - \tan^2 30^\circ$ is: 1

(a) $\frac{3}{2}$ **(b)** $\frac{13}{6}$ **(c)** $\frac{7}{3}$ **(d)** $\frac{7}{6}$

43. A statue 14 m tall stands on a pedestal. From a point on the ground, the angle of elevation of the top of the statue is 60° and that of the top of the pedestal is 45°. The height of the pedestal is: 1

(a) $\frac{14\sqrt{3}}{3}$ m **(b)** $7 + 7\sqrt{3}$ m **(c)** $14\sqrt{3}$ m **(d)** 14 m

44. PA and PB are tangents from an external point P to a circle, and $\angle PAB = 35^\circ$. The $\angle APB$ is: 1

(a) 55° **(b)** 110° **(c)** 70° **(d)** 35°

45. A wire in the form of a circle of radius 21 cm is bent into a square. The area of the square is: 1

(a) 1089 cm² **(b)** 33 cm² **(c)** 4356 cm² **(d)** 1386 cm²

46. A toy is a cone of base radius 42 cm and height 56 cm mounted on a hemisphere of the same radius (the flat faces coincide). The total surface area of the toy is: 1

(a) 20328 cm² **(b)** 9240 cm² **(c)** 25872 cm² **(d)** 18480 cm²

47. If $\sum f_i = 40$ and $\sum f_i x_i = 1200$, the mean is: 1

(a) 1160 **(b)** 300 **(c)** 30 **(d)** 40

48. A letter is chosen at random from the word 'PERMUTATION'. The probability that it is a vowel is: 1

(a) $\frac{4}{11}$ **(b)** $\frac{5}{11}$ **(c)** $\frac{5}{12}$ **(d)** $\frac{6}{11}$

49. The LCM of 10, 15 and 20 is: 1

(a) 30 **(b)** 60 **(c)** 70 **(d)** 120

50. If α, β are zeroes of $x^2 - 5x + 4$, then $\frac{1}{\alpha} + \frac{1}{\beta}$ is: 1

(a) $\frac{4}{5}$ **(b)** $-\frac{5}{4}$ **(c)** $\frac{5}{4}$ **(d)** 20

END OF QUESTION PAPER **Total: 50 Marks**

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